

# The ReACT Agent Loop (Per-Turn Reason + Act + Observe)

**Scope.** This document describes the **core ReACT agent loop** — the per-turn loop that runs every time you send a message to ragent. It is *not* the goal-driven /loop feature (see TUI-QUICKSTART.md, section 5, for /loop). Where the two interact, the interaction is called out explicitly.

## 1. What it is

Every user message triggers an agentic **Reason -> Act -> Observe** cycle:

1. **Reason** — the session processor streams an LLM response for the current conversation state.
2. **Act** — if the model requests tool calls, each call is permission-checked and executed.
3. **Observe** — the tool results are appended to the conversation history and fed back into the next LLM request.

The cycle repeats until the model produces an answer **without** any tool calls, or a termination condition fires (step limit, cancel, interrupt, or an unrecoverable error).

| Element                                            | Location                                                                                                                                                                                                                                                    |
|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Orchestrator struct                                | SessionProcessor —<br>crates/ragent-agent/src/session/processor.rs:106                                                                                                                                                                                      |
| Public entry (plain text prompt)                   | process_message — processor.rs:1137                                                                                                                                                                                                                         |
| Public entry (pre-built Message, multipart/images) | process_user_message —<br>processor.rs:1164                                                                                                                                                                                                                 |
| <b>The core loop</b>                               | inline loop { ... } — processor.rs:1472,<br>inside process_user_message                                                                                                                                                                                     |
| Per-step helpers                                   | crates/ragent-agent/src/session/loop_steps.rs<br>— prepare_client (:121),<br>build_turn_system_prompt (:362),<br>build_turn_chat_messages (:649),<br>run_inline_init_acknowledgement (:739),<br>call_llm_step (:857),<br>finalize_assistant_message (:1529) |
| Loop-run state machine                             | LoopSpec / LoopTracker / StopCondition —<br>crates/ragent-agent/src/session/loop_state.rs                                                                                                                                                                   |

SessionProcessor orchestrates the loop: it accepts a user message, streams an LLM response, executes any requested tool calls, and iterates until the model signals completion or the step limit is reached.

## 2. Turn lifecycle

### Phase A — turn setup (before the loop)

Location: processor.rs:1171-1465.

1. **Telemetry** — an `agent_loop_profiler()` and a `SessionRecorder` are created and `record_session_start()` is logged.
2. **User message persisted** — `storage_op(move |s| s.create_message(&msg))` writes the message to SQLite on a blocking thread; `Event::MessageStart` is published.
3. **RunId issued** — `RunId::new()`; the message is recorded in the activity log as a user model message. This `run_id` links every event of the turn.
4. **Client/model/config resolution** — `prepare_client` resolves the provider client, model reference, session config, working directory and team context. Failure here is an unrecoverable provider-stage error: the run is terminated with status error and `Err` is returned.
5. **Per-run cost tracking** — a `TokenUsage` listener accumulates (`input`, `output`) tokens for the turn; when the turn ends a `RunCostSummary` row is computed from merged price entries and published as `Event::RunCostSummary`.
6. **System prompt build** — `build_turn_system_prompt` assembles the agent preset, initiatives, tool reference, codeindex guidance and team guidance via the cached `SystemPromptCache`.
7. **Chat history build** — `build_turn_chat_messages` returns the provider chat messages, whether compression happened, the last reported input token count, and the context window. The vector is shared behind an `Arc<Vec<ChatMessage>>` so per-retry requests share it by refcount.
8. **AGENTS.md init acknowledgement** — `run_inline_init_acknowledgement` prints a one-time display-only acknowledgement (skipped for sub-agents).
9. **Loop setup:**
  - `max_steps = agent.max_steps.unwrap_or(1024)` — the default iteration ceiling is **1024** unless the agent preset sets one.
  - `event_bus.set_step(session_id, 0)`.
  - Tool definitions are chosen: **empty if `max_steps <= 1`** (single-shot: the model gets no tools); filtered to the active `/loop` tool set when one is active; otherwise the full cached tool registry.
  - A reusable assistant placeholder `Message` is created and persisted — its id anchors all streaming updates to one DB row.
  - Interrupt plumbing: `interrupt_requested = user cancel flag OR active /loop interrupt flag`.

## Phase B — one ReACT iteration

Each pass of the loop { body (`processor.rs:1472+`) runs this guard chain in order:

| # | Guard                | Behaviour                                                                                                                                                                                                                      |
|---|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Stop-flag guard      | If the loop tracker <code>is_stopped()</code> , break — no stage runs.                                                                                                                                                         |
| 2 | Safe-point interrupt | On <code>/loop</code> runs, a user cancel <b>or</b> a raised <code>/loop</code> interrupt breaks <b>before</b> the budget gate (first stop wins; user intervention takes precedence) and ends the run with status interrupted. |

| # | Guard             | Behaviour                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 | Budget gate       | <code>tracker.begin_step()</code> returns false when the <code>/loop</code> step or token budget was reached; the run ends as <code>budget_exhausted</code> <b>before another LLM request is sent.</b>                                                                                                                                                                                                                           |
| 4 | Step counter      | <code>set_step(current + 1)</code> under the <code>loop.step.total</code> profile scope.                                                                                                                                                                                                                                                                                                                                         |
| 5 | Max-steps check   | <code>step &gt; max_steps -&gt; warning +</code><br><code>Event::AgentError("Reached maximum steps (N)") +</code><br><code>break.</code>                                                                                                                                                                                                                                                                                         |
| 6 | User-cancel check | On plain (non-loop) turns, a set cancel flag logs the timing breakdown, persists the partial assistant message, publishes <code>Event::MessageEnd { reason: Cancelled }</code> , records <code>TerminationReason::Interrupted</code> , publishes the run-cost summary, and returns an empty message. On <code>/loop</code> runs this branch is effectively plain-turn-only — a loop-run cancel already ended the run at guard 2. |

Then the reasoning stage:

7. **Event::ToolsSent** — published **only on step 1 and only when tools are offered** (`!tool_definitions.is_empty()`), so the TUI renders the offered tool list once.
8. **Pre-send auto-compaction** — when `compaction.auto` is enabled and not yet attempted this turn, the request is token-estimated and `evaluate_trigger` runs against the context window; if the trigger fires, history is summarised (`crate::compaction::compact`), a synthetic `Role::Compaction` message is persisted, and the in-memory history is swapped to compaction-summary + verbatim recent tail. Compaction failure only warns and continues uncompressed.
9. **LLM call** — `call_llm_step` builds the `ChatRequest` (system prompt, history, tool definitions) and folds the provider stream (`futures::StreamExt`) into incremental `text_buffer`, `reasoning_buffer` and accumulated `tool_calls`. A fatal stream error terminates the run with status error.
10. **Provider-reported input tokens** — persisted into session state so the TUI context panel shows the same figure next turn.
11. **Token tally** — `tracker.record_tokens(...)` is persisted; the *next* iteration's budget gate sees the updated totals.
12. **Response parts** — non-empty reasoning becomes a `MessagePart::Reasoning` on the assistant

message; non-empty text publishes `Event::ModelResponse` (preview, elapsed ms, token counts) and becomes a `MessagePart::Text`.

### Phase C — decision point: tool call vs final answer

```
if llm_result.tool_calls.is_empty()
  -> the text response IS the final answer; the loop ends
else
  -> tool dispatch phase; afterwards `continue` back to Phase B
```

When the model answers **without** tool calls, three sub-paths run (in order):

- **Sub-agent summary nudge** — if this is a sub-agent run past step 1 whose text is under the 2000-byte narration limit, the narration was likely “Now let me check...” rather than findings. The narration is popped, pushed back into history as an assistant message, and a `SUBAGENT_SUMMARY_NUDGE` user message requests the real summary; the loop continues. This happens **once per run** and is *not* a termination.
- **Verification gate** (/loop only, T-005) — when a spec with a `verify_cmd` is active, `run_verification_command` decides: **pass** -> the run ends as `GoalAchieved` with the `verify` label; **fail with steps remaining** -> the failure output becomes the next observation and the loop continues (a success record resets the consecutive-failure counter so unrelated recoverable failures do not burn the retry allowance); **fail with budget breached** -> ends as `budget_exhausted`.
- **Plain stop** — a no-tool-call response with no verification command means the goal was reached -> run ends as `GoalAchieved`.

When tool calls are **non-empty**, one more interrupt safe point runs, then the **tool dispatch phase** (`processor.rs:2042+`): text is moved (not cloned) into a `ContentPart::Text`, `parallel_tool_calls` is read from `experimental.parallel_tool_calls`, one `ToolContext` is built per step, and per tool call: **permission check** -> **execution** -> **result collection** -> **interim save**, then continue back to the top of the loop so the next LLM request carries the observations.

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## 3. Flow diagram

```
User message
|
v
persist message -> RunId issued -> prepare_client (fatal on error)
|
v
build system prompt + chat history (cached)
|
+-----+
| loop { (processor.rs:1472)                                |
|   stop-flag? -> break                                     |
|   /loop interrupt (or cancel on loop runs)?               |
|   -> status interrupted, break                             |
|   budget breach? -> budget_exhausted, break               |
|   step > max_steps? -> AgentError, break                   |
| }                                                         |
+-----+
```

```

|   plain-turn cancel? -> persist partial,           |
|       MessageEnd{Cancelled}                       |
|   ToolsSent (step 1 only)                         |
|   auto-compaction (if enabled, once per turn)     |
|   call_llm_step <-- REASON (stream fold)          |
|   record tokens / response parts                  |
|                                                    |
|   tool_calls empty? --yes--> final answer, end    |
|       |no                                         |
|       v                                         |
|   interrupt safe point                           |
|   per tool: permission -> execute -> observe      |
|   append tool results -> continue loop            |
+-----+

```

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#### 4. Tool calls vs final answer

The decision is purely structural: a streamed response that accumulates one or more `tool_calls` is an **action** turn; a response with **no** tool calls is the **final answer** and terminates the loop. Thinking/reasoning blocks are buffered separately (`reasoning_buffer`) and persisted as `MessagePart::Reasoning` — they never terminate the loop on their own.

Tool-call arguments arrive incrementally; assembly is managed through `PendingToolCall` (`history.rs`) before a call is promoted into `llm_result.tool_calls` and dispatched.

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#### 5. Permissions and tool execution

Tool execution is gated by the multi-layered permission system:

- `PermissionChecker` consults global rules (permission array in `ragent.json`) and per-agent rules (`agent.<name>.permission`).
- Allow/Deny verdicts are honoured; an Allow that would auto-approve can still be forced into an interactive prompt when a `/loop spec` with `checkpoints: true` is active (destructive tools are checkpointed).
- `--yes` / `--no-prompt` (CLI) or `/yolo` (TUI) auto-approve every prompt; explicit Deny rules still take precedence over allow rules.
- A permission prompt times out to **deny** (safe default).

A tool task that neither finishes nor panics is killed at the tool watchdog timeout (`TOOL_WATCHDOG_TIMEOUT` = 2000 s) and the whole run is terminated with status `error`; an ordinary task join failure is logged and skipped instead.

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#### 6. Streaming, stalls, retries

- **Stream driver:** `call_llm_step` folds `StreamEvents` into the per-step working state (`LoopState`): incremental text, reasoning, tool calls.

- **Thinking blocks** are configured per provider/model (see Section 8, thinking) and persisted as `MessagePart::Reasoning`.
- **Stall detection** — `stream.timeout_secs` bounds the gap between subsequent stream deltas; a stall triggers the retry path. Stalls are detected via `stall_pattern_set()` with `should_flush`.
- **Retry/backoff** — `stream.max_retries` attempts with linear backoff ( $N * \text{stream.retry\_backoff\_secs}$ ). Retryability helpers: `should_retry_stream_error`, `stream_has_meaningful_partial_output`, `is_token_overflow_error_message`, `is_permanent_llm_api_error`.
- **Initial response timeout** — `stream.initial_response_timeout_secs` waits for the first byte (RTT + provider cold start).

UI event fan-out during a turn: `MessageStart` -> step counter -> `ToolsSent` (step 1) -> `ModelResponse` -> per-tool `ToolCallArgs` / `ToolCallEnd` / `ToolCallStart` (inside the spawned tool task; `ToolCallBatch` at step end) -> `MessageEnd`.

## 7. Termination conditions

| # | Condition                                           | Signal / status                                                                                                          |
|---|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 1 | Model answers without tool calls                    | <code>GoalAchieved</code> (plain turns simply return the message)                                                        |
| 2 | /loop verification gate passes                      | <code>GoalAchieved</code> with the verify label                                                                          |
| 3 | Step budget exhausted                               | <code>AgentError</code> "Reached maximum steps (N)"; default ceiling 1024 (agent override) / <code>--maxsteps</code> cap |
| 4 | /loop token cost budget exhausted                   | <code>BudgetExhausted</code> , fired before the next request                                                             |
| 5 | User cancel                                         | <code>FinishReason::Cancelled</code> , partial message persisted                                                         |
| 6 | Human interrupt (Esc during a /loop run)            | <code>HumanIntervention</code> -> status interrupted; takes precedence over budget at safe points                        |
| 7 | Unrecoverable provider error (prep or stream stage) | status error, <code>Err</code> returned                                                                                  |
| 8 | Tool watchdog                                       | tool stalls past 2000 s -> run terminated                                                                                |

Not terminations: the one-shot sub-agent summary nudge, and verification failures that still have steps remaining (they become the next observation).

## 8. Configuration options

ragent reads configuration from `ragent.json` (or `ragent.jsonc`) with the following precedence, each later layer merged over the previous:

1. Compiled defaults
2. Global file: `~/.config/ragent/ragent.json`
3. Project file: `./.ragent/ragent.json` (created with defaults if neither file exists)
4. `RAGENT_CONFIG` env var -> file path overlay
5. `RAGENT_CONFIG_CONTENT` env var -> inline JSON overlay

Per-provider blocks and per-model entries are deep-merged, so partial overrides preserve lower layers. Resolved config is cached keyed on the mtime + size of every contributing file plus the cwd (`CachedConfigFile`); every `Config::save()` invalidates that cache so same-process writes are immediately visible to subsequent loads.

### Core loop knobs

| Key                                                | Type                                   | Default                            | Behaviour                                                          |
|----------------------------------------------------|----------------------------------------|------------------------------------|--------------------------------------------------------------------|
| <code>defaultAgent</code>                          | String                                 | "general"                          | Default agent preset driving the loop.                             |
| <code>agent.&lt;name&gt;.max_steps</code>          | <code>Option&lt;u32&gt;</code>         | unset (falls back to <b>1024</b> ) | Iteration ceiling for the loop when this agent is active.          |
| <code>agent.&lt;name&gt;.model</code>              | String                                 | unset                              | <code>provider:model</code> override for this agent.               |
| <code>agent.&lt;name&gt;.temperature</code>        | <code>f32</code>                       | unset                              | Sampling temperature.                                              |
| <code>agent.&lt;name&gt;.top_p</code>              | <code>f32</code>                       | unset                              | Nucleus sampling.                                                  |
| <code>agent.&lt;name&gt;.permissions</code>        | rules                                  | []                                 | Per-agent permission rules.                                        |
| <code>permission</code>                            | <code>Vec&lt;PermissionRule&gt;</code> | []                                 | Global allow/deny/ask rules (permission, pattern, action).         |
| <code>yolo</code>                                  | bool                                   | false                              | Trust mode: auto-approve everything.                               |
| <code>experimental.parallel_tool_calls</code>      | bool                                   | false                              | Execute requested tool calls in parallel rather than sequentially. |
| <code>experimental.max_background_agents</code>    | usize                                  | 8                                  | Concurrent background sub-agent cap.                               |
| <code>experimental.background_agent_timeout</code> | u64                                    | 3600                               | Sub-agent timeout (seconds).                                       |

### agent\_perf — per-step performance tuning

Consulted at loop startup and on every step. `validate()` refuses to start on violations.

| Key                             | Type | Default | Behaviour                                                    |
|---------------------------------|------|---------|--------------------------------------------------------------|
| <code>agent_perf.enabled</code> | bool | true    | Master switch; false short-circuits every perf optimisation. |

| Key                                   | Type    | Default                       | Behaviour                                                          |
|---------------------------------------|---------|-------------------------------|--------------------------------------------------------------------|
| agent_perf.profiling                  | bool    | false                         | Per-scope timing logs at info level.                               |
| agent_perf.step_budget                | u64secs | 300                           | Max wall-clock seconds per agent step ( $\geq 5$ ).                |
| agent_perf.stall_timeout              | u64secs | 60                            | Seconds without a stream delta before stall recovery ( $\geq 5$ ). |
| agent_perf.max_concurrent_tools       | u32     | min(available_parallelism, 4) | Max parallel tool calls per turn ( $\geq 1$ ).                     |
| agent_perf.parallel_independent_tools | bool    | true                          | Execute independent tool calls in parallel.                        |

#### stream — request timeouts and retries

| Key                                  | Type | Default | Behaviour                                                       |
|--------------------------------------|------|---------|-----------------------------------------------------------------|
| stream.initial_response_timeout_secs | u64  | 300     | Wait for the <b>first byte</b> .                                |
| stream.timeout_secs                  | u64  | 120     | Max gap between <b>subsequent</b> deltas before stall -> retry. |
| stream.max_retries                   | u32  | 4       | Retry attempts after stall/connection failure ( $\leq 32$ ).    |
| stream.retry_backoff_secs            | u64  | 2       | Attempt N waits N * <code>retry_backoff_secs</code> .           |

Validation: both timeouts  $\geq 5$ ; `initial_response_timeout_secs`  $\geq$  `timeout_secs`.

#### compaction — history compaction

| Key                    | Type | Default | Behaviour                                                          |
|------------------------|------|---------|--------------------------------------------------------------------|
| compaction.auto        | bool | true    | Auto-compact before a send when the trigger fires (once per turn). |
| compaction.threshold   | f32  | 0.7     | Fraction of context window that triggers compaction.               |
| compaction.buffer      | f32  | 0.10    | Headroom kept after compaction.                                    |
| compaction.keep.tokens | f32  | 0.20    | Fraction of recent tokens kept verbatim after compaction.          |



## thinking — reasoning blocks

Configured per provider and per model; model-level overrides provider-level.

| Key                                              | Type    | Default            | Behaviour                                |
|--------------------------------------------------|---------|--------------------|------------------------------------------|
| provider.<id>.thinking.enabled                   | boolean | provider-dependent | Enable thinking blocks for the provider. |
| provider.<id>.thinking.effort                    | String  | provider-dependent | Reasoning effort (e.g. low/medium/high). |
| provider.<id>.models.<id>.thinking.budget.tokens | int     | provider-dependent | Token budget for reasoning.              |

## Tool visibility

tool\_visibility switches hide tool families from tool definitions and prompt listings (the tools stay registered and executable):

| Key                         | Default |
|-----------------------------|---------|
| tool_visibility.office      | false   |
| tool_visibility.github      | false   |
| tool_visibility.gitlab      | false   |
| tool_visibility.teams       | false   |
| tool_visibility.agents      | false   |
| tool_visibility.plan        | false   |
| tool_visibility.codeindex   | true    |
| tool_visibility.masterfetch | true    |
| tool_visibility.browser     | true    |
| tool_visibility.finance     | true    |

## The loop section — /loop only, not the core loop

The loop section (loop.max\_steps, loop.cost\_limit, loop.error\_retry\_allowance, loop.checkpoints, loop.checkpoint\_timeout\_secs) configures the **goal-driven /loop feature**, which injects stop conditions and a verification gate into the core loop at its safe points. It does not change how a plain per-turn turn runs.

## 9. Commands

### CLI (src/main.rs)

| Flag / command            | Effect                                           |
|---------------------------|--------------------------------------------------|
| --model <MODEL>           | Override provider/model for the loop.            |
| --agent <AGENT>           | Agent preset driving the loop (default general). |
| --yes (alias --no-prompt) | Auto-approve all permission prompts.             |
| --no-tui                  | Headless stdout instead of the TUI.              |

| Flag / command                                      | Effect                                                                                                                                                         |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--maxsteps &lt;N&gt;</code>                   | Cap agentic loop steps (CLI help text says 500, but with no agent override the effective ceiling is the agent preset's <code>max_steps</code> — default 1024). |
| <code>--no-git-context / --no-readme-context</code> | Disable git/README context injection into the system prompt.                                                                                                   |
| <code>--dry-run</code>                              | Readiness check only; never invokes model or tool.                                                                                                             |
| <code>ragent run &lt;PROMPT&gt;</code>              | One-shot agent execution of a single prompt.                                                                                                                   |
| <code>ragent serve [--addr &lt;ADDR&gt;]</code>     | Start the HTTP server (default 127.0.0.1:3000).                                                                                                                |
| <code>ragent session resume &lt;ID&gt;</code>       | Resume a session into a live loop.                                                                                                                             |

### TUI slash commands

| Command                                         | Effect on the loop                                                          |
|-------------------------------------------------|-----------------------------------------------------------------------------|
| <code>/agent [&lt;name&gt;]</code>              | Switch the agent preset driving the loop.                                   |
| <code>/model [&lt;id&gt;]</code>                | Switch model mid-session.                                                   |
| <code>/thinking</code>                          | Configure thinking mode.                                                    |
| <code>/tools</code>                             | Tool registry / visibility control.                                         |
| <code>/compact</code>                           | Force context compaction of loop history.                                   |
| <code>/cost</code>                              | Show the run-cost summary.                                                  |
| <code>/cancel</code>                            | Cancel in-flight processing.                                                |
| <code>/clear</code>                             | Clear the session transcript.                                               |
| <code>/system</code>                            | Show the current system prompt.                                             |
| <code>/undo</code>                              | Roll back to the last file snapshot.                                        |
| <code>/yolo</code>                              | Toggle YOLO trust mode (auto-approve).                                      |
| <code>/autopilot on [--max-tokens N]</code>     | Autonomous loop continuation with token/time budget.                        |
| <code>[--max-time N] / off / status</code>      |                                                                             |
| <code>/reload</code>                            | Hot-reload subsystems into the running loop.                                |
| <code>[agents\ config\ mcp\ skills\ all]</code> |                                                                             |
| <code>/context refresh</code>                   | Clear the prompt-context cache (file tree/git/README recomputed next turn). |

### HTTP API (Bearer token required; GET /health is open)

| Endpoint                                | Shape                                                             |
|-----------------------------------------|-------------------------------------------------------------------|
| POST /sessions/{id}/messages            | Body {"content": String} — the primary loop-driving endpoint.     |
| POST /sessions/{id}/abort               | Aborts the running loop.                                          |
| POST /sessions/{id}/permission/{req_id} | Body {"decision": "allow" "always" "deny"}.                       |
| GET /events                             | SSE event stream (MessageStart, ModelResponse, ToolCallEnd, ...). |
| POST /sessions/{id}/tasks               | Spawn a sub-agent (agent, task, background?, model?).             |

## Keyboard shortcuts

- **Esc** during processing — request a loop interrupt (“stopping after the current stage”).
  - **Ctrl+C** — copy-to-clipboard when text selection is active, else quit arm; **Ctrl+D** confirms quit.
- 

## 10. Worked examples

### One-shot CLI run

```
export ANTHROPIC_API_KEY="sk-..."
ragent run --agent coder "Refactor src/main.rs to split the CLI wiring"
```

### Driving the loop over HTTP

```
# create a session
SID=$(curl -s -X POST http://127.0.0.1:3000/sessions \
  -H "Authorization: Bearer $RAGENT_TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"directory": "."}' | jq -r .id)

# drive the loop
curl -s -X POST http://localhost:3000/sessions/$SID/messages \
  -H "Authorization: Bearer $RAGENT_TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"content": "List the files in crates/ and summarise each crate"}'

# watch events
curl -N http://localhost:3000/events -H "Authorization: Bearer $RAGENT_TOKEN"
```

### Autopilot continuation

```
/autopilot on --max-tokens 50000 --max-time 600
```

The loop continues autonomously until the task completes, the budgets are exhausted, or /autopilot off is issued.

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## 11. Observability

- **Run cost** — every turn publishes `Event::RunCostSummary` with accumulated input/output tokens and cost derived from merged prices entries (`PriceEntry { model, input_per_1m, output_per_1m }`, USD per 1M tokens).
  - **Activity log** — append-only JSONL event store keyed by `RunId`; every user/assistant message and termination reason is recorded.
  - **LLM request stats** — `llmstats slash` command shows request statistics (counts, durations, token tallies).
  - **OpenTelemetry** — `telemetry` section (or legacy `experimental.openTelemetry`) enables OTLP export of session spans.
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## 12. Related documentation

- `TUI-QUICKSTART.md` — using the terminal UI (section 5 covers `/loop`)
- `QUICKSTART.md` — installation and first-run configuration
- `SPEC.md` — full configuration schema
- `docs/howtos/loopprogramming.md` — the goal-driven `/loop` extension that wraps this loop with a spec and stop conditions
- `docs/howtos/office.md` — the office + PDF tool families this loop can execute (`tool_visibility.office` gated)
- `docs/howtos/teams.md` — multi-agent team coordination
- `docs/performance/benchmark-guide.md` — benchmarking the loop surfaces